25900386 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFSQ200 - FOOD PRESERVATION AND ADDITIVES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Describe one limitation of sterilization in food preservation. [U] / [CO1]

   a). Loss of some nutrients and sensory qualities

   b). No effect on microbes

   c). Too cheap to apply

   d). Makes food raw

2. Identify the typical pressure range used in HPP for food preservation. [K] / [CO2]

   a). 10–50 MPa

   b). 100–200 MPa

   c). 300–600 MPa

   d). 1000–2000 MPa

3. The main principle of concentration is [U] / [CO1]

   a). Increase in water activity

   b). Reduction of water activity and volume

c). Addition of salt and sugar d). Chemical sterilization

4. The expansion valve in a refrigeration cycle functions to [U] / [CO1]

a). Increase refrigerant pressure b). Decrease refrigerant pressure

c). Convert liquid into solid d). Compress refrigerant vapor

5. The material commonly used for electrodes in ohmic heating system is: [U] / [CO2]

a). Glass b). Stain less steel

c). Wood d). Plastic

6. Select major limitation of canning is ____. [U] / [CO1]

a). Short shelf life b). High cost and nutrient loss

c). High fat requirement d). Low microbial safety

7. Select the component of food that absorbs most microwave energy. [K] / [CO2]

a). Proteins b). Carbohydrates

c). Water molecules d). Fats

8. Enzyme inactivation in foods by heat is important to prevent ____. [U] / [CO1]

a). Ripening and spoilage b). Fermentation

c). Freezing damage d). Caramelization

9. Name the permitted additive would you apply in soups to enhance umami flavor? [A] / [CO3]

a). Pectin b). Aspartame

c). Monosodium glutamate d). Citric acid

10. To balance acidity in fruit juices, which additive would be applied? [A] / [CO3]

a). Antioxidants b). Acidulants

c). Colorants d). Sweeteners

[1x10 = 10]

Part B

Short Answer

Answer any **4** questions

Each question carries **5** marks

11.	Classify the types of pasteurization based on temperature and time.	[U] / [CO1]
12.	Describe the different types of membranes used in food applications.	[U] / [CO2]
13.	Discuss the effect of food additives on human health.	[U] / [CO3]
14.	Interpret the role of temperature in food preservation.	[U] / [CO1]
15.	A food label shows E330. Apply your knowledge to identify the additive and explain its role in foods.	[A] / [CO3]
16.	Explain physical methods of sterilization	[U] / [CO1]

[5x4 = 20]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

17.	Elaborate the principle, process, equipment and application of food irradiation in preservation	[U] / [CO2]
18.	Explain the importance of antioxidants and their classification in food processing.	[A] / [CO3]
19.	Describe different types of microbial spoilage.	[K] / [CO1]
20.	Explain the reasons behind the food dehydration	[U] / [CO1]

[10x2 = 20]